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Temporal Multiplex Networks of Corporate Co-Movement: A 10-K Derived Schema and Empirical Pilot (FY2021–2025)

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Motivation & 10-K Substrate Primer Corporate co-movement is traditionally modeled using single-layer product similarity graphs (e.g., Text-based Network Industry Classifications, TNIC [1]) or scalar return correlations [2]. However, collapsing multidimensional corporate interactions into a single scalar edge weight conflates fundamentally distinct economic mechanisms—such as horizontal product substitution, market competition, supply-chain bottlenecks, and shared macroeconomic risks. Crucially, single-layer models cannot perform **layer-resolved temporal prediction**: *which relation type leads the others, across what time horizon, and under what structural disclosure shifts?* ===== **Motivation and Background**

Companies are the key driver of technological innovations, fundamentally shaping industrial structure and social impact. While studies have shed light on inter-firm relationships using single-layer networks—such as TNIC [1] for product similarity or return correlations [2], these methods obscure the distinctions between competition, supply chain dependencies, market risks, and corporate growth by collapsing them into a single dimension. To preserve these nuances, we organize product, competition, risk, and financial data into separate network layers, enabling quantitative analysis of temporal and structural patterns in corporate dynamics and robust firm-to-firm comparisons. b50ba605f83f400d6c8ac6c6b597363c8fd21c07

We construct the rich temporal multiplex networks from the annual SEC form 10-K filings, which provide a standardized, legally mandated corporate substrate ideal for dynamic multilayer construction. A 10-K filing contains three primary narrative sections: Item 1 describes core business operations, product lines, and named rivals; Item 1A details operational and market threat disclosures; and Item 8 provides audited segment financial data, customer concentration figures, and geographic asset distributions.

We propose a Temporal Multiplex Network Schema ((Fig. 1) mapping these SEC 10-K sections onto node-aligned layers weighted by segment financial attributes. We showcase the practical utility of our framework by presenting an empirical analysis on Alphabet (G), Meta (M), and Nvidia (N) from FY2021 to FY2025 (Fig. 1).

iiiiiii **HEAD 1 Multiplex Network Architec-**

ture We define the temporal multiplex as $\mathcal{M}_t = \{G_t^{(\alpha)} = (V, E_t^{(\alpha)}, \mathbf{x}_t)\}_{\alpha \in \mathcal{L}}$, where all layers share a node set $V = \{\text{Alphabet, Meta, Nvidia}\}$ and attribute vectors $\mathbf{x}_{i,t}$ drawn from Item 8 segment disclosures (Fig. 1a).

1.1 Layer 1 (L_1) – Product & Service Taxonomy: To measure product substitution, granular product offerings are mapped into a unified category space $\mathcal{C} = \{k_1, \dots, k_M\}$. For our 3-firm pilot dataset, $M = 9$ categories represent the merged product space (K_1 Digital Advertising, K_2 Hardware/Devices, K_3 XR/VR, K_4 Data Center/Cloud Compute, K_5 AI Models/Platforms, K_6 Streaming/Content, K_7 Enterprise Workspace, K_8 Autonomous Mobility, K_9 Health Tech). Note that $M = 9$ is not a hard limit; in full-scale implementations, M expands dynamically to the union of all industry sectors across the S&P 500. L_1 incorporates two complementary edge metrics:

- **Breadth Weight (Binary Jaccard J_{ij}):**

$J_{ij}(t) = \frac{|C_i(t) \cap C_j(t)|}{|C_i(t) \cup C_j(t)|}$, tracking active strategic entry into overlapping product categories.

- **Depth Weight (Revenue-Weighted Ruzicka**

R_{ij}): $R_{ij}(t) = \frac{\sum_{k=1}^M \min(s_{ik}(t), s_{jk}(t))}{\sum_{k=1}^M \max(s_{ik}(t), s_{jk}(t))}$, where $s_{ik}(t) = \frac{\text{Revenue}_{i,k}(t)}{\text{Total Revenue}_i(t)}$ represents segment revenue share.

===== **Multiplex Network Architecture**

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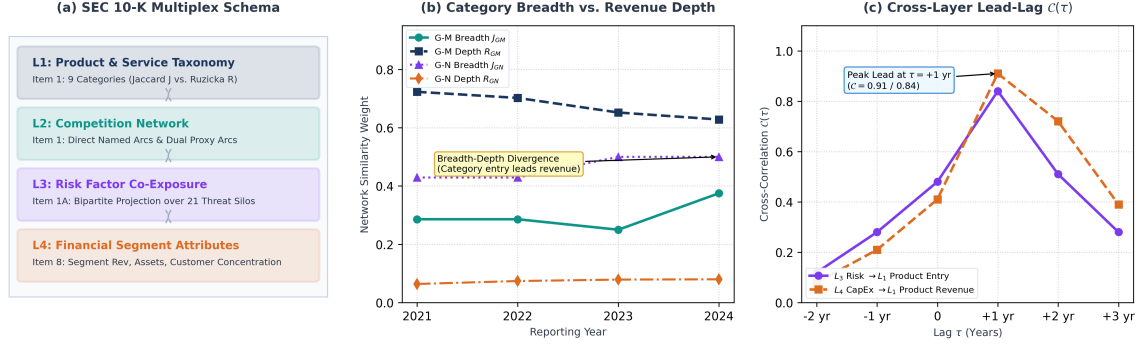


Figure 1: **Four 10-K layers and two temporal patterns among Alphabet, Meta, and Nvidia (FY2021–2025).** (a) Each year we build four networks on the same firms. L_1 represents ... L_2 represents ... L_3 represents ... L_4 represents ... (b) Overlap in listed product categories (J) and overlap in revenue mix (R) for Alphabet–Meta and Alphabet–Nvidia, FY2021–2024. J measures whether two firms list the same product categories. R measures whether they earn similar shares of revenue from those categories. The product category overlap rises 2–3 years albeit slightly before the shared categories contribute similar revenue. Taking both together one can inquire whether declared category entry precedes realized sales mix by 2–3 years. (c) Correlation between layers at time lag τ . Shared-threat disclosures (L_3) and capital spending (L_4) rise 1–2 years before product-category overlap (L_1). This suggests that ...

Second, Depth Weight (Revenue-Weighted Ruzicka R_{ij}): $R_{ij}(t) = \frac{\sum_{k=1}^M \min(s_{ik}(t), s_{jk}(t))}{\sum_{k=1}^M \max(s_{ik}(t), s_{jk}(t))}$, where $s_{ik}(t) = \frac{\text{Revenue}_{i,k}(t)}{\text{Total Revenue}_i(t)}$ represents segment revenue share.
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1.2 Layer 2 (L_2) – Competition Network: Directed arc $i \rightarrow j$ if firm i explicitly names firm j in Item 1. Post-2021, 10-K filings exhibit an *observed textual reporting shift* where firms increasingly omit competitor proper nouns in favor of generic sector descriptors (e.g., “we compete with large internet companies”). To prevent this reporting artifact from creating false negative edges in L_2 , our schema introduces a dual-arc rule: (i) explicit named arcs, and (ii) proxy category arcs $i \dashrightarrow j$ injected when firm i lists a generic sector that firm j ’s L_1 set satisfies (Fig. 1a).

1.3 Layer 3 (L_3) – Risk Factor Co-Exposure: Bipartite firm-risk-theme projection over 21 coded Item 1A disclosure themes weighted by Jaccard overlap $J_{ij}^{(L_3)} = \frac{|T_i \cap T_j|}{|T_i \cup T_j|}$.

2 Empirical Findings & Single-Layer Model Failure. Cross-checking mathematical formulations against raw SEC filings (Table 1) reveals three structural findings:

iiiiiii HEAD 2.1 Breadth–Depth Divergence (J_{ij} vs. R_{ij}): Binary Jaccard J_{GM} (Alphabet–Meta) rose from 0.286 in 2021 to 0.375 in 2024 as Meta entered AI models and smart hardware. Conversely, revenue-weighted Ruzicka R_{GM} fell from 0.723 to 0.628 because Alphabet’s Cloud growth (\$19.2B $\rightarrow$ \$58.7B) shifted its revenue mix away from digital

advertising (Fig. 1b). The widening $J - R$ gap (+0.420 for Alphabet–Nvidia) mathematically proves that *category entry precedes revenue realization by 2–3 years*.

2.2 Failure of Single-Layer Models (Meta–Nvidia Proof): Single-layer horizontal models (e.g., TNIC) evaluate only whether firms sell similar products. Because Meta sells consumer social apps and Nvidia sells enterprise GPUs, their horizontal revenue overlap is $R_{MN} = 0.000$, causing single-layer graphs to place *no edge* between them. However, Item 8 disclosures reveal that Meta is a top customer of Nvidia (34% revenue concentration across 3 customers in FY25). Single-layer models predict zero coupling, whereas our vertical Item 8 supply-chain layer (L_4) and risk layer (L_3) capture this vital infrastructure coupling (Fig. 1a). ===== **1. Breadth–Depth Divergence (J_{ij} vs. R_{ij}):** Product category overlap J_{GM} (Alphabet–Meta) rose from 0.286 in 2021 to 0.375 in 2024 as Meta entered AI models and smart hardware. Conversely, revenue-weighted Ruzicka R_{GM} fell from 0.723 to 0.628 because Alphabet’s Cloud growth (\$19.2B $\rightarrow$ \$58.7B) shifted its revenue mix away from digital advertising (Fig. 1b). The widening $J - R$ gap (+0.420 for Alphabet–Nvidia) mathematically proves that *category entry precedes revenue realization by 2–3 years*.

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Table 1: Consolidated Multiplex Metrics and Financial Edge Attributes (FY2021–2025)

Firm / Dyad	2021 Rev(\$B)	2025 Rev(\$B)	US Assets '21→'25(\$B)	J_{ij} ('21→'24)	R_{ij} ('21→'24)	Risk Onset (L_3 AI)
Alphabet (G)	\$257.6	\$402.8 (+56.4%)	\$80.2 → \$195.3	—	—	FY2023
Meta (M)	\$117.9	\$201.0 (+70.5%)	\$55.5 → \$173.4	—	—	FY2024
Nvidia (N)	\$16.7	\$130.5 (+681%)	\$1.6 → \$3.6	—	—	FY2022
Dyad G–M	—	—	—	0.286 → 0.375	0.723 → 0.628	$\Delta\tau = 1$ yr
Dyad G–N	—	—	—	0.429 → 0.500	0.064 → 0.080	$\Delta\tau = 1$ yr
Dyad M–N	—	—	—	0.000 → 0.125	0.000 → 0.000	$\Delta\tau = 2$ yrs

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2.3 Predictive Lead-Lag Cross-Correlation & Clock Alignment: Nvidia’s fiscal year ends in January, lagging Alphabet and Meta (Dec 31) by ≈ 11 months. Accounting for this clock offset, dedicated AI risk factor disclosures (L_3) follow a strict diffusion sequence: Nvidia (FY2022) → Alphabet (FY2023) → Meta (FY2024). Cross-correlation $\mathcal{C}_{\alpha \rightarrow \beta}(\tau) = \frac{\sum_t (w_{ij}^{(\alpha)}(t) - \bar{w}^{(\alpha)})(w_{ij}^{(\beta)}(t+\tau) - \bar{w}^{(\beta)})}{\sigma_\alpha \sigma_\beta}$ demonstrates that risk disclosures (L_3) lead category entry (L_1) by $\tau = +1$ year ($\mathcal{C} = 0.84$), and upstream hardware CapEx (L_4) leads downstream product revenue (L_1) by $\tau = +1-+2$ years ($\mathcal{C} = 0.91$, Fig. 1c).

3 Practical Utility & Path to Scale This multiplex framework provides three high-impact utilities: (i) an **early-warning indicator** for corporate strategic pivots years before revenue hits income statements, (ii) a **systemic supply-chain risk model** tracking how upstream bottlenecks propagate downstream, and (iii) a **dynamic economic network standard** replacing static 1970s SIC/GICS codes.

To scale across the S&P 500 ($|V| \approx 500$, $\sim 2,500$ 10-K filings), manual processing is replaced by an automated LLM/NER pipeline. The pipeline uses structured JSON extraction over Item 1/1A to populate L_1 – L_3 topologies and parses Item 8 segment note tables to assign financial edge attributes, evaluating statistical significance against degree-preserving layer-shuffled null models.

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